

Experiment No.04

PART A

A.1 Aim: Identify the Classification problem and create a Knowledge database for that problem and apply appropriate search method for optimization.

A.2 Prerequisite: Understand Knowledge Database, Machine Learning, Data Mining

A.3 Outcome:

After successful completion of this experiment students will be able to

□ To create knowledge base and apply appropriate problem solving method for optimization.

A.4 Theory:

1. Identify classification problem and Create Data set for the same problem as a knowledge database.
2. Explore the Data set using any machine learning tool e.g WEKA, R-programming.
3. Select one of the Relevant Classification algorithm for your selected data set.
4. Identify relevant attribute by applying any of the three Attribute Relevant Methods and appropriate search technique as listed below and shown in following Figure A.
 - 1) CFSEvaluator
 - 2) ChiSquared Attribute Evaluator
 - 3) Attribute SubsetEvaluator etc.
5. Analyse the data set and find out Performance Evaluation Matrix as shown in the following Figure B.
6. Compare Performance parameters for above three Attribute selection Methods.

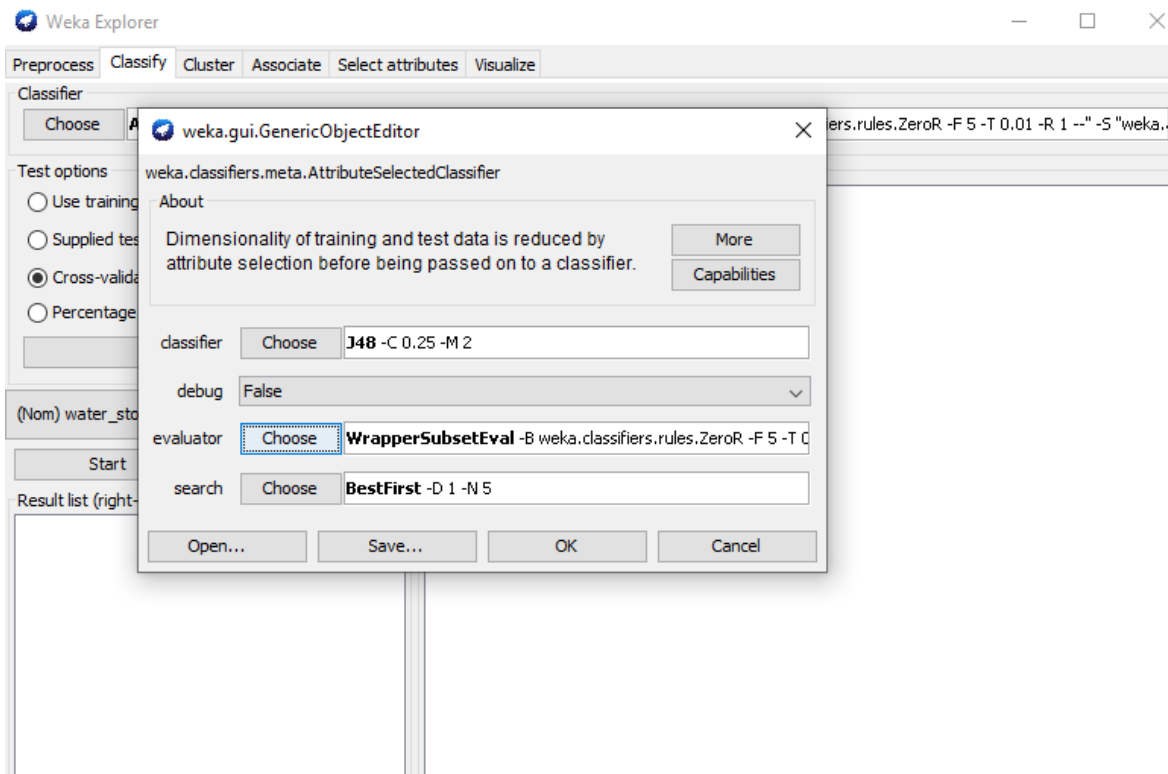


Figure: A .Selection of Attribute Selector method in WEKA.

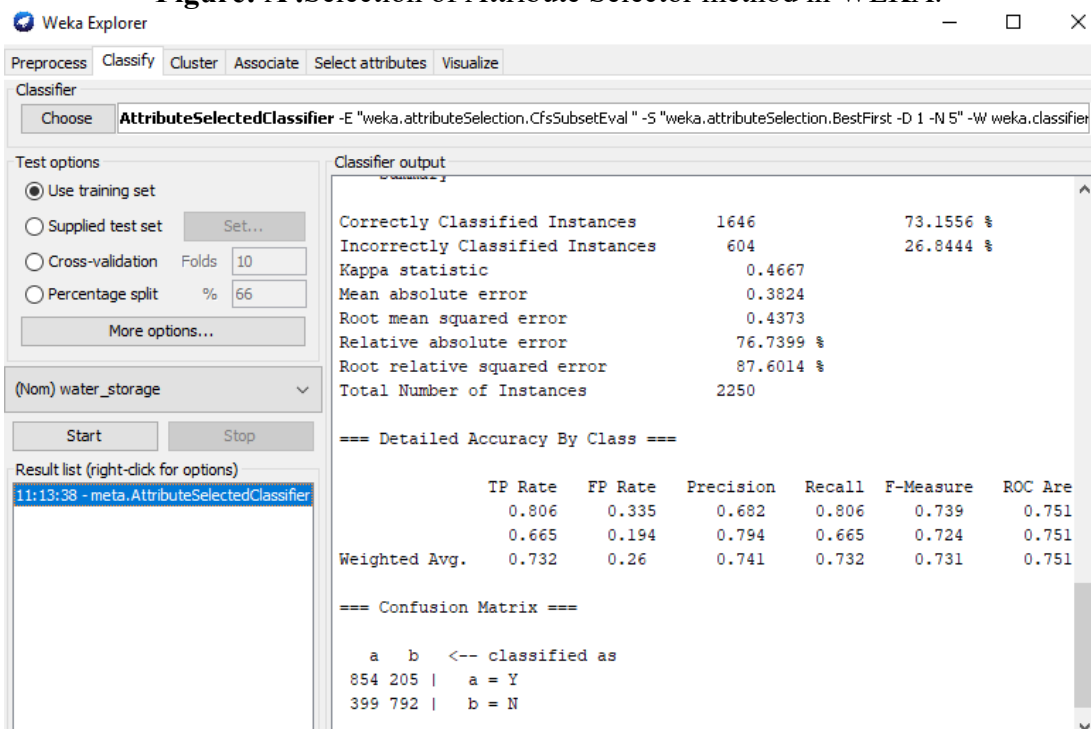


Figure B.Performance Evaluation Matrix in WEKA.

PART B

(PART B: TO BE COMPLETED BY STUDENTS)

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Class: TE Comps C	Batch: C-3
Date of Experiment: 28.01.25	Date of Submission:5.02.25
Grade:	

B.1 Software Code written by student:

@RELATION supermarket

@ATTRIBUTE customer_id NUMERIC

@ATTRIBUTE age {young, middle-aged, senior}

@ATTRIBUTE gender {male, female}

@ATTRIBUTE income {low, medium, high}

@ATTRIBUTE product_category {grocery, electronics, clothing, household, other}

@ATTRIBUTE purchase_amount NUMERIC

@ATTRIBUTE payment_method {cash, card, digital_wallet}

@ATTRIBUTE loyalty_program {yes, no}

@DATA

1, young, male, medium, grocery, 25.50, card, yes

2, middle-aged, female, high, electronics, 300.00, digital_wallet, no

3, senior, female, low, grocery, 15.20, cash, yes

4, young, male, high, clothing, 120.00, card, no

5, middle-aged, male, medium, household, 80.75, digital_wallet, yes

6, senior, female, medium, other, 40.00, cash, no

7, young, female, high, grocery, 30.00, card, yes

8, middle-aged, female, low, clothing, 45.00, cash, no

9, senior, male, medium, electronics, 200.00, digital_wallet, yes

10, young, male, low, other, 10.50, cash, no

B.2 Input and Output:

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply Stop

Current relation
Relation: supermarket
Instances: 4627

Attributes: 217
Sum of weights: 4627

Attributes: All None Invert Pattern

No.	Name
1	☐ department1
2	☒ department2
3	☐ department3
4	☐ department4
5	☐ department5
6	☐ department6
7	☐ department7
8	☐ department8
9	☐ department9
10	☐ grocery misc
11	☐ department11
12	☐ baby needs
13	☐ bread and cake
14	☐ baking needs
15	☐ coupons
16	☐ juice-sat-cord-ms
17	☐ tea
18	☐ biscuits
19	☐ canned fish-meat

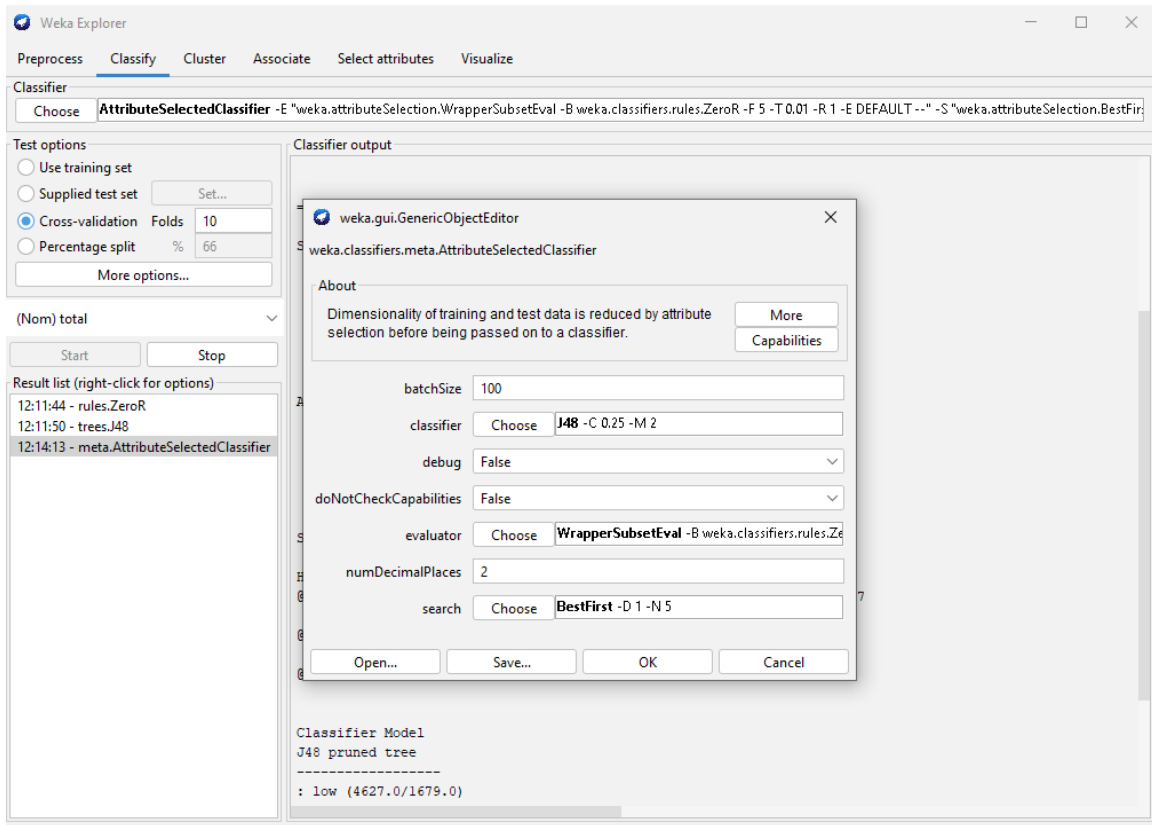
Remove

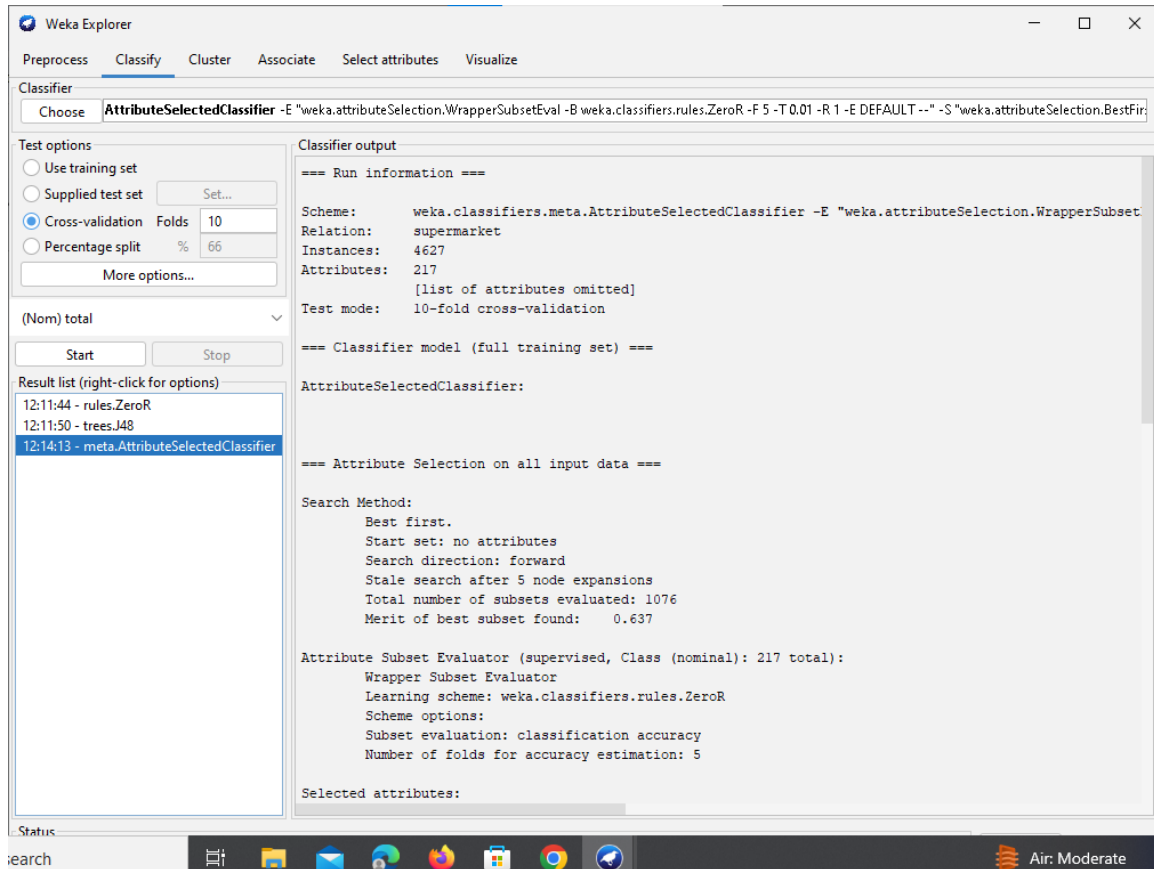
Selected attribute
Name: department2
Missing: 4496 (97%)
Distinct: 1
Type: Nominal
Unique: 0 (0%)

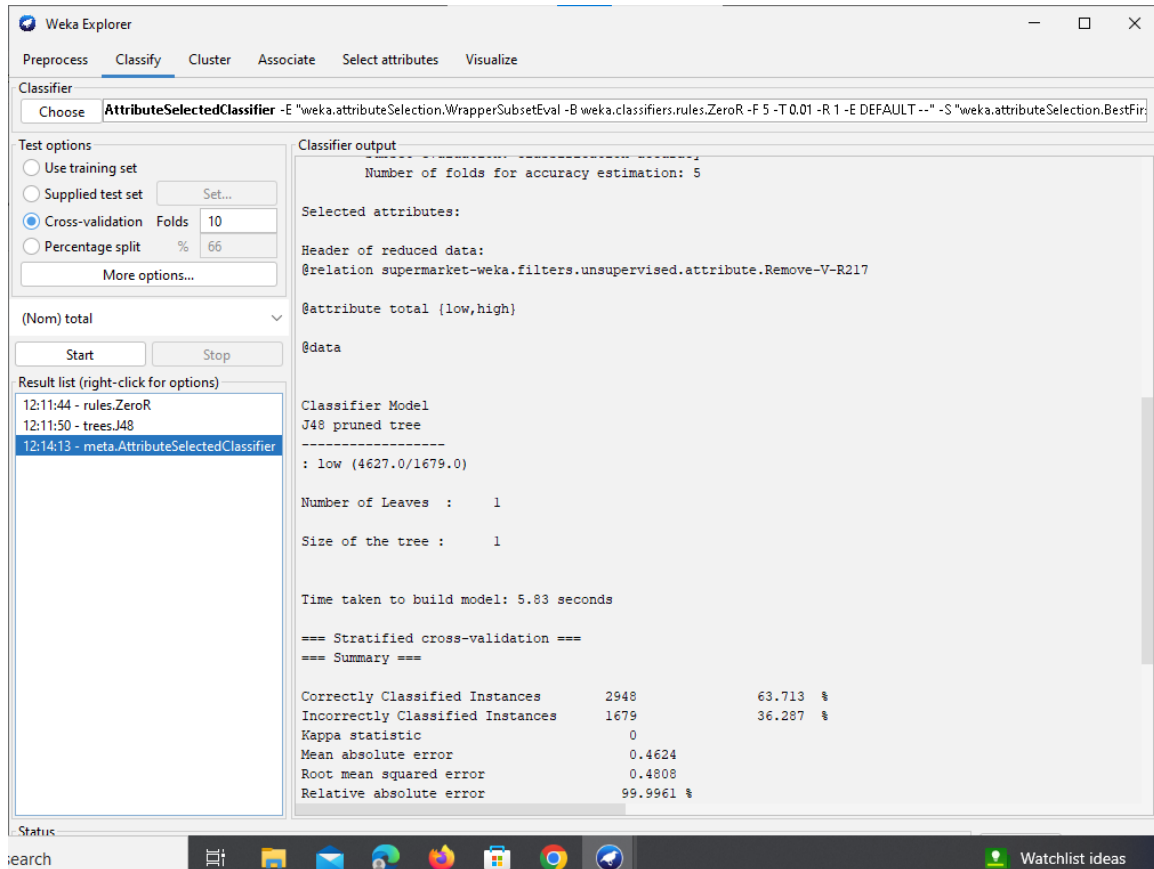
No.	Label	Count	Weight
1	t	131	131

Class: total (Nom) Visualize All

Status







Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **AttributeSelectedClassifier** -E "weka.attributeSelection.WrapperSubsetEval -B weka.classifiers.rules.ZeroR -F 5 -T 0.01 -R 1 -E DEFAULT --" -S "weka.attributeSelection.BestFirst"

Test options

- ☐ Use training set
- ☐ Supplied test set
- ☒ Cross-validation Folds: 10
- ☐ Percentage split %: 66

More options...

(Nom) total

Start Stop

Result list (right-click for options)

- 12:11:44 - rules.ZeroR
- 12:11:50 - trees.J48
- 12:14:13 - meta.AttributeSelectedClassifier

Classifier output

```

Number of Leaves : 1
Size of the tree : 1

Time taken to build model: 5.83 seconds

=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances 2948 63.713 %
Incorrectly Classified Instances 1679 36.287 %
Kappa statistic 0
Mean absolute error 0.4624
Root mean squared error 0.4808
Relative absolute error 99.9961 %
Root relative squared error 100 %
Total Number of Instances 4627

=== Detailed Accuracy By Class ===

```

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
low	1.000	1.000	0.637	1.000	0.778	?	0.499	0.637	low
high	0.000	0.000	?	0.000	?	?	0.499	0.363	high
Weighted Avg.	0.637	0.637	?	0.637	?	?	0.499	0.537	

=== Confusion Matrix ===

```

a b <-- classified as
2948 0 | a = low
1679 0 | b = high

```

Status: OK

chi-square test

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **AttributeSelectedClassifier** -E "weka.attributeSelection.ChiSquaredAttributeEval" -S "weka.attributeSelection.Ranker" -T 1.7976931348623157E308 -N 1 -W weka.classifiers.trees.J48 --C 0.25 -M 2

Test options

- ☐ Use training set
- ☐ Supplied test set
- ☒ Cross-validation Folds: 10
- ☐ Percentage split %: 66

More options...

(Nom) total

Start Stop

Result list (right-click for options)

- 11:37:33 - trees.J48
- 11:37:56 - trees.J48
- 11:38:36 - meta.AttributeSelectedClassifier
- 11:38:56 - meta.AttributeSelectedClassifier

Classifier output

```

@attribute department145 {t}
@attribute total {low,high}

@data

Classifier Model
J48 pruned tree

: low (4627.0/1679.0)

Number of Leaves : 1
Size of the tree : 1

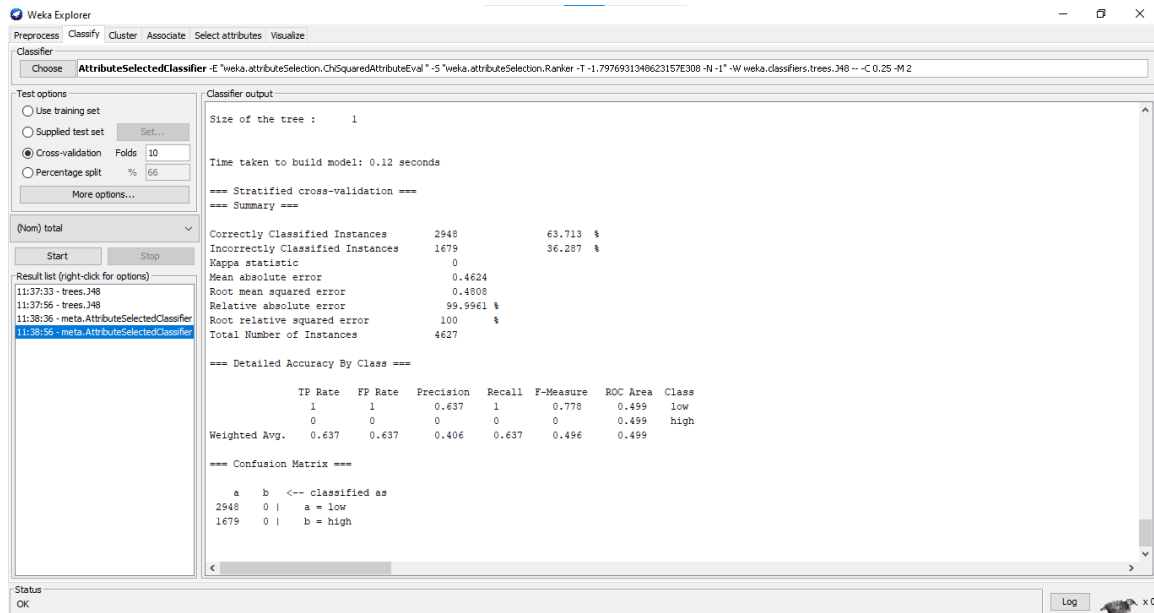
Time taken to build model: 0.12 seconds

=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances 2948 63.713 %
Incorrectly Classified Instances 1679 36.287 %
Kappa statistic 0
Mean absolute error 0.4624
Root mean squared error 0.4808
Relative absolute error 99.9961 %
Root relative squared error 100 %
Total Number of Instances 4627

=== Detailed Accuracy By Class ===

```

Status: OK



B.3 Observations and learning: (Performance Evaluation)

After running J48 with optimized parameters, we evaluate the model's accuracy, precision, recall, and F1 score on the test set. If the accuracy is satisfactory, the model can be deployed for classifying new emails.

A decision tree created by J48 will be interpretable, allowing us to visualize the decision-making process. This could be valuable for understanding why an email is classified as spam or not.

B.4 Conclusion:

Using the J48 algorithm to solve the spam classification problem has provided us with an interpretable decision tree model. By applying optimization techniques such as Grid Search or Random Search, we have fine-tuned the model's hyperparameters to achieve better performance. This method successfully identifies spam emails with a clear decision-making process, making it a practical solution for this classification task.

B.5 Question of Curiosity

(To be answered by student based on the practical performed and learning/observations)

Q1: What are the different methods for Relevant Attribute Selection?

Methods for Relevant Attribute Selection

1. **Filter Methods:** Rank features based on statistical measures (e.g., Chi-Square, Information Gain, Correlation).
2. **Wrapper Methods:** Evaluate feature subsets using a learning algorithm (e.g., Recursive Feature Elimination).
3. **Embedded Methods:** Select features during model training (e.g., Lasso Regression, Decision Trees like J48).
4. **Dimensionality Reduction:** Transform features into a lower-dimensional space (e.g., PCA, LDA).
5. **Genetic Algorithms:** Use evolutionary techniques to select optimal feature subsets.

Q2: Explain Performance Evaluation Parameters for Classification Problem.

Performance Evaluation Parameters for Classification

1. **Accuracy:** Proportion of correct predictions.
2. **Precision:** $\text{Correct positive predictions} / \text{Total predicted positives}$.
3. **Recall:** $\text{Correct positive predictions} / \text{Actual positives}$.
4. **F1 Score:** Harmonic mean of precision and recall.
5. **Specificity:** $\text{Correct negative predictions} / \text{Actual negatives}$.
6. **AUC-ROC:** Area under the ROC curve, evaluating classifier performance across thresholds.
7. **Confusion Matrix:** Displays true positives, false positives, true negatives, and false negatives.
8. **Log Loss:** Penalizes wrong predictions based on predicted probabilities.